

DECODELABS

Data Analytics Internship
Complete Project Report



Prepared by: Isaac Okolie
Program: DecodeLabs | SkillAhead Solutions Ltd
Accreditation: Institute of Analytics (IoA), UK
Year: 2026

PROJECT 1	PROJECT 2	PROJECT 3	PROJECT 4
Data Cleaning	EDA	SQL Analysis	Power BI

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1. Executive Summary

This report documents the complete work completed during the DecodeLabs Data Analytics Internship — a program powered by SkillAhead Solutions Ltd and accredited by the Institute of Analytics (IoA), United Kingdom. The internship involved four end-to-end data analytics projects built on a real e-commerce sales dataset containing 1,200 orders across 2023–2025.

The projects covered the full data analytics workflow: data cleaning, exploratory analysis, SQL querying, and Power BI dashboard design. Each project was supported by a formal written report, change log documentation, and professional deliverables suitable for stakeholder review.

\$1,264,762	1,200	18	4
Total Revenue Analyzed	Total Orders	SQL Queries Written	Projects Completed

2. Internship Overview

Property	Detail
Program	DecodeLabs Data Analytics Internship
Powered by	SkillAhead Solutions Ltd
Accreditation	Institute of Analytics (IoA), United Kingdom

Property	Detail
Intern	Isaac Okolie
Year	2026
Focus	Real-world data analytics using industry tools

Tools & Technologies Used

Tool	Purpose
Microsoft Excel	Data cleaning, EDA, pivot tables, charts
Power Query (M Language)	Data transformation and shaping
MySQL	SQL querying and data analysis
Microsoft Power BI	Data visualization and dashboard design
DAX	Measures and calculated columns in Power BI

3. Dataset Overview

All 4 projects were built on the same e-commerce sales dataset. This ensured consistency across the full analytics workflow — from raw data to final dashboard.

Property	Detail
Dataset Name	E-Commerce Orders Table
Total Records	1,200 Orders
Total Customers	1,189 Unique Customers
Total Products	7
Total Revenue	\$1,264,762
Analysis Period	2023 – 2025
Total Columns	14

Key Fields:

OrderID, CustomerID, Product, Quantity, UnitPrice, TotalPrice, PaymentMethod, ReferralSource, CouponCode, OrderStatus, TrackingNumber, ShippingAddress, ItemsInCart, Date

PROJECT 1**Data Integrity & Cleaning Audit****Excel & Power Query**

Objective

Audit and clean the raw dataset before any analysis. Identify all data quality issues, resolve them appropriately, and document every change in a formal Change Log.

Key Work Done

- Inspected 1,200 rows using Power Query Column Distribution and Column Quality tools.
- Identified 309 missing values in the CouponCode column (~30% of dataset).
- Investigated missing values — confirmed they indicated no coupon was used, not broken data.
- Filled missing CouponCode values with 'No Coupon' preserving all 309 records.
- Converted the Date column from numeric format to proper date using Locale settings.
- Verified OrderID has zero duplicates (1,000 distinct = 1,000 unique).
- Confirmed CustomerID duplicates are expected — one customer can place multiple orders.
- Applied IQR method to flag outliers in UnitPrice and TotalPrice columns.
- Investigated all 8 TotalPrice outliers — confirmed as valid high-volume orders.

Change Log

Change ID	Description	Impact	Status
CR001	Replaced 309 null CouponCode values with 'No Coupon'	309 records preserved	Resolved
CR002	Converted Date column from numeric to date format using Locale	All rows reformatted	Resolved
CR003	Flagged UnitPrice outliers using IQR — Upper Fence \$1,024.83	0 outliers detected	Verified
CR004	Flagged 8 TotalPrice outliers — investigated, retained as valid high-volume orders	0 records removed	Resolved

Core Principle: If it isn't documented, it didn't happen.

PROJECT 2	Exploratory Data Analysis (EDA)	Excel & Power Query
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Objective

Explore the cleaned dataset to uncover patterns, detect outliers, measure relationships between variables, and communicate findings through visualizations.

Descriptive Statistics

Metric	UnitPrice	TotalPrice	Quantity	ItemsInCart
Mean	\$356.41	\$1,053.97	2.95	5.49
Median	\$364.00	\$824.00	3	5
Min	\$11.39	\$11.39	1	1
Max	\$699.93	\$3,456.40	5	10

IQR Outlier Detection

Column	Q1	Q3	IQR	Upper Fence	Outliers Found
UnitPrice	\$186.06	\$521.57	\$335.51	\$1,024.83	0
TotalPrice	\$410.52	\$1,578.48	\$1,167.96	\$3,330.41	8 (all valid)

Correlation Analysis

Column Pair	Correlation (r)	Strength
UnitPrice → TotalPrice	0.72	Strong Positive
Quantity → TotalPrice	0.62	Moderate-Strong
Quantity → ItemsInCart	0.65	Moderate-Strong
UnitPrice → ItemsInCart	0.00	No Relationship

Visualizations Built

Chart	Type	Key Finding
Total Orders by Product	Bar Chart	Tablet & Printer lead in volume
Order Status Count	Column Chart	Cancelled orders lead all statuses
Orders by Referral Source	Bar Chart	Instagram dominates with 259 orders
Revenue by Referral Source	Column Chart	Instagram generates \$275,285
Revenue by Product	Bar Chart	Tablet leads at \$186,569
Revenue by Payment Method	Column Chart	Payments evenly distributed
Monthly Revenue Trend	Line Chart	Consistent downward trend 2023–2025

PROJECT 3	SQL for Data Analysis	MySQL
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Objective

Use structured SQL queries to extract actionable business insights from the e-commerce dataset across revenue, customers, products, marketing, and order fulfillment.

Database Setup

```
CREATE DATABASE Project_3; | USE Project_3;
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Query Summary

#	Query	Category	Key Result
1-3	View, count & preview dataset	Inspection	1,200 records confirmed
4-5	Unique values & fix CouponCode nulls	Cleaning	309 nulls replaced
6	Total revenue	Revenue	\$1,264,738
7	Average order value	Revenue	\$1,053.95
8	Unique customers	Customer	1,189 customers

#	Query	Category	Key Result
9-10	Revenue & quantity by product	Product	Varied by category
11	Revenue by payment method	Payment	Near-equal distribution
12	Revenue by referral source	Marketing	Instagram leads
13	Orders by status	Fulfillment	Cancelled = 250 orders
14	Revenue lost to returns	Fulfillment	\$243,268
15	Revenue lost to cancellations	Fulfillment	\$276,390
16	Avg cart size by referral source	Behavior	Varies by channel
17	Most used coupon code	Promotional	Usage pattern identified
18	Revenue by year	Trend	Year-on-year breakdown

Key Financial Results

\$1,264,738	\$1,053.95	\$276,390	\$243,268
Total Revenue	Avg Order Value	Lost to Cancellations	Lost to Returns

PROJECT 4	Data Visualization & Storytelling	Power BI
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Objective

Transform the analyzed data into a professional 2-page Power BI dashboard that communicates revenue performance and revenue leakage through clear visual storytelling.

Design Principles Applied

Principle	Application
One Page, One Message	Page 1 = Revenue Drivers Page 2 = Revenue Leakage
Reduce Visual Clutter	Only visuals directly supporting the business question included
Tell a Story	Every visual answers: What happened? Why? What next?

Page 1 — Revenue Driver Analysis

\$1,264,762	\$275,285	\$195,620	5.49
Total Revenue	Instagram Revenue	Top Product (Chair)	Avg Items in Cart

Referral Source	Revenue
Instagram	\$275,285 — #1
Email	\$261,809 — #2
Google	\$250,441 — #3
Facebook	\$250,411 — #4
Referral	\$226,816 — #5

Page 2 — Revenue Leakage Analysis

\$519,674	\$276,396	\$243,278	41.4%
Total Revenue Lost	Lost to Cancellations	Lost to Returns	Combined Loss Rate

8. Key Business Insights Across All Projects

#	Insight	Source
1	Total revenue of \$1,264,762 generated across 2023–2025	Projects 2, 3, 4
2	Instagram is the highest-performing acquisition channel at \$275,285	Projects 3, 4
3	Chair, Printer, and Laptop are the top 3 revenue-generating products	Projects 3, 4
4	41.4% of all orders either cancelled or returned — the biggest operational challenge	Projects 2, 3, 4
5	\$519,674 in revenue was lost and is potentially recoverable	Projects 3, 4
6	Revenue shows a consistent downward trend since mid-2024	Projects 2, 4
7	UnitPrice is the strongest driver of TotalPrice (r = 0.72)	Project 2

#	Insight	Source
8	All 5 payment methods are near-equally used — no single method dominates	Project 3
9	309 CouponCode records were missing — filled without data loss	Projects 1, 3
10	TotalPrice outliers were valid high-volume orders, not errors	Projects 1, 2

9. Skills Demonstrated

Skill Area	Evidence
Data Cleaning & Auditing	Identified and resolved 309 missing values; IQR outlier detection
Exploratory Data Analysis	Descriptive stats, distribution analysis, correlation analysis
Statistical Analysis	Mean, Median, IQR, Pearson Correlation Coefficient
SQL & Database Management	18 queries across 6 analytical categories in MySQL
Data Visualization	7 Excel charts + 2-page Power BI dashboard
Business Communication	4 formal reports with insights and recommendations
Documentation	Change logs, README files, project reports for all 4 projects
Storytelling with Data	Every visual designed to answer a specific business question

10. Final Conclusion

This internship provided hands-on experience with the complete data analytics workflow — from dirty raw data to professional business dashboards. Each project built on the previous one, reinforcing the principle that clean data, rigorous analysis, and clear communication are all equally important in delivering real business value.

The four projects collectively demonstrate the ability to: identify and resolve data quality issues, extract insights using multiple tools, detect patterns and anomalies, and communicate findings in a way that supports executive decision-making.

The most significant business finding across all projects is the \$519,674 in recoverable revenue currently being lost through cancellations and returns — representing the single greatest opportunity for immediate

financial improvement without additional acquisition spend.

Sign-Off

Prepared by: Isaac Okolie

Role: DecodeLabs Data Analytics Intern

Date: 2026

Raw data. Real analysis. Actionable insights.